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Interventions to Improve Breast Screening Uptake for Black African Women in High Income Countries: A Systematic Review

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Abstract

Aim Black African women experience significant barriers and are less likely to attend breast screening. However, little is known about the interventions and the components of successful interventions for Black African women. The aim of this systematic review was to identify and evaluate interventions developed for Black African women to increase breast screening in high-income countries.

Methods A systematic review was conducted using nine databases (AMED, the British nursing database, CINAHL, COCHRANE, EMBASE, MEDLINE, PSYCHINFO, Scopus and Web of Science). These databases yielded 1767 studies which were imported into Covidence and screened. 12 studies were included. Interventions were mapped to the behaviour change technique taxonomy (BCT) and evaluated using direction and magnitude of effect.

Results All included studies were based in the United States (US). Interventions include health promotional messages through radio and print media, individual support and community support programs. 11 out of 12 studies showed that interventions were effective in increasing mammogram uptake, screening knowledge or mammogram intention. Individual support or patient navigation program with a problem-solving BCT appeared to be the most promising intervention and provides robust guidance to screening.

Conclusion There were few studies which were all US based, suggesting that other countries have not developed interventions for this population group. Thus, population specific interventions should be explored in other countries. Most studies addressed individual barriers, there is a need for studies to address provider and health system barriers. Interventions with problem-solving approach should be explored in other countries.

Keywords Breast cancer screening · Mammogram · Health interventions · Black African women · Systematic review

Background

Black African women are less likely to attend breast cancer screening than other ethnic groups of women. In the United Kingdom (UK) for instance, despite breast screening being routinely available free of charge for all women aged 50–70 years, Black African women have low screening attendance rates [1]. There was 11% decrease in odds of attendance of screening per 10% increase in black population groups [2]. This means that, with increasing populations of black women, there is low attendance of breast screening, when compared to other population groups. It is noteworthy that there are significant differences with the Black ethnic group category. A subpopulation analysis of breast screening amongst Black women in the US highlighted that Black African women are less likely to attend breast screening [3]. Similar results were obtained in the UK, when screening

The authors have read the journal policies and is submitting this manuscript in accordance with those policies.

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attendance was compared between Black Caribbean ethnic group and Black African women [4]. Furthermore, Black Caribbean women (63%) were more likely to attend first invitations than Black African women (49%); while for routine screening recalls, Black Caribbean women (74%) were more likely to take up the invitation than Black African women (64%) [5].

To further compound the problems associated with late screening or low screening uptake, Black African women are prone to developing aggressive type of breast cancer like the triple negative breast cancer (TNBC) which can easily metastasize and even reoccur; and are at high risk of having BRCA 1 and 2 hereditary genes, that can potentially be passed on to their children [6–8]. For instance, a study assessing the prevalence of BRCA 1 and 2 genes reported that, 1 in 7 women of Black African women who had genetic testing had BRCA mutation, independent of age and family history [9]. Gathani et al., comparing ethnicity and tumour characteristics of invasive breast cancer has reported that Black African women have significantly greater adjusted odds of several of the less favourable tumour characteristics, than their white counterparts [10]. It has also been shown that TNBC in populations of African ancestry populations is high in Ghanaians and others with West African ancestry origin [11].

The majority of Black African women in high income countries are immigrants or have immigrant parents who are still new to the health system of the host country [12]. Black African women come from different African countries, which has been acknowledged as one of the most sociocultural diverse continents in the world. There are social, cultural, religious, and language differences within these African countries, with the populations' health care behaviour in High income counties likely being influenced by the knowledge or experience of health system in their place of origin and the length of residence in their host country [13]. Most African health systems focus on the curative rather than a preventative breast screening approach [14] while high income countries emphasize on both preventive and curative approaches to health care [15]. According to the World Health Organisation breast screening in 88% of African countries, follow an opportunistic or a non-organised approach, the methods vary and not standardized [16]. There are also healthcare related norms from back home which are practiced by Black African women. For instance amongst East African women, being generally healthy and symptom free was a barrier to screening [17]. In Ethiopia, health service providers do not integrate health promotion and disease prevention services but rather focus on curative health services [14]. Kundi analysed the impact of public health awareness on preventive healthcare and found that knowledge has significant impact on preventive approach to health [18]. And so, women migrating from these countries will likely lack awareness of mammogram and preventive healthcare programs, therefore unlikely to attend screening when there is no obvious health concern. Interestingly, a New York study highlighted that the most endorsed barrier is lack of insurance and being treated rudely at mammogram centre [19] indicating provider and system level barriers. Acculturation and integration which is the process by which immigrants embrace the attitudes, behaviours and beliefs of their host country [20], could help shape the attitude towards health care utilization and possibly overcome these barriers, but this could potentially take a life time in some cases. Breast screening offered to women can help identify breast cancer early, helps to ensure prompt treatment, and increases chances of survival.

There is an evidence gap, as most reviews exploring interventions to increase breast screening uptake focus on the general population or other ethnic groups more generally [21–27] or focused on a specific intervention [28–31] and did not highlight the impact of interventions on screening utilization for Black African women. A previous systematic review by Nwadinigwe and colleagues established that most studies from high income countries, do not highlight barriers or facilitators for screening attendance, specific for this population group [32], therefore there is need to identify successful interventions to increase breast screening in Black African women in high income countries like the UK. Consequently, this systematic review will explore what interventions have been used with Black African women to encourage screening uptake, and which of these interventions were effective. Public health interventions are complex, thus it is useful to analyse the effect of different components present or absent in an intervention [33]. Hence, the Behaviour Change Technique (BCT) taxonomy which classified 93 techniques to change behaviour was adopted. The BCT is a component or an active ingredient within an intervention designed to change a specified behaviour [34]. The BCT taxonomy was used as it provides precision in identifying the effective components of an intervention and promotes future replication [35]. The primary aim of this review was to identify and evaluate the intervention developed to improve breast screening service uptake for Black African women in high income countries.

Methods

The systematic review was registered on PROSPERO (CRD42024510472) and followed the Preferred reporting items systematic reviews and meta-analyses guidelines [36].

Inclusion/Exclusion Criteria

Our review aimed to identify and evaluate interventions that has been developed to increase breast screening service uptake amongst Black African women in High income countries. To capture all studies on interventions relevant to this population group, the PICO framework (population, intervention, comparator and outcome) was used to formulate the aim of the review, due to its sensitivity and for a comprehensive search [37].

- Population-Black African women.
- Intervention-community-based education or patient navigation programs or any intervention.
- Comparator-usual care or no intervention or any intervention used for comparative purposes.
- Outcomes-mammography uptake, screening knowledge, intention to screen or any other outcomes.

All primary study types from High income countries, that have been peer-reviewed, published in English and focusing on interventions for Black African women were included. Studies with other ethnic groups were included, only if results from Black African women could be extracted. The year limitation was from 2012 to 2025. This followed implementation of the World Health Organisation action plan for developing cost-effective interventions against major non-communicable disease, including population-based breast screening [38]. The World Bank's definition of High-Income country was adopted [39].

Search Strategy

A range of relevant databases that relate to health care issues were identified. These were Web of Science, Scopus, Medline, CINAHL, Cochrane, British Nursing database, Psychinfo, AMED and Embase. A sample of the search strategy is provided in Table 1 below. The Cochrane library was searched as it provides systematic reviews that can be hand searched to obtain relevant studies. Grey literature was not searched as we wanted to focus on high quality, scientific, peer reviewed journal articles to ensure the quality of the methodology used by the authors. The search strategy is presented in supplementary files.

Table 1 Showing search strategy

S1) mammogram* or mammograph*	
S2) breast* N4 (screen* or "health examin*" or "health check" or investigat* or examin*)	S1 and S2 combined with Boolean term OR. Now called S5
S3) (black* or Africa*) N4 (women or woman or female*)	S3 and S5 combined with Boolean term AND. Now called S6
S4) Intervention* or strateg* or program* or plan* or scheme* or project* or initiative*	S4 and S6 combined with Boolean term AND. Now called S7

Screening Process

A total of 1767 studies were imported into Covidence, none of the systematic reviews that were retrieved from Cochrane and hand searched, yielded relevant studies. The results were imported into Covidence library, 709 duplicates were removed, leaving the reviewer with 1058 titles and abstracts to screen. 132 studies were retrieved and their full text assessed for eligibility. A total of 1755 studies were excluded. The reason for their exclusion is highlighted in the PRISMA diagram below in Fig. 1. A total of 12 studies were included. Data extraction was done using Microsoft Word tables. The search strategy, the screening process and data extraction was carried out by the first author (UN) and the quality of this process independently checked by the second author (II). Discrepancies between authors were resolved through discussion. Study characteristics, participant demographics, characteristics of intervention and outcomes were extracted.

Risk of Bias Assessment

This was done using ROB 2 [40] and ROBINS-I tool [41] for randomized control trials (RCTs) and non-randomised control trials respectively. The ROB 2 tool uses signalling question to assess bias from the randomization process, deviation from intended intervention, missing data, measurement of outcomes and selection outcome. For ROB2, a study is said to have low risk of bias if it is judged to be low risk in all assessment domains. A study was deemed to have some concerns if there is at least one domain with some concerns, and at high risk if one domain in the assessment has a high-risk score. Similarly, the ROBINS-I tool, assesses bias from confounding factors, classification of interventions, deviation from intended interventions, missing data and selection bias. Overall, a low risk of bias score is given when all domains have low risk or have some concern about uncontrolled confounding factor. To be judged as moderate risk, a study must have at least one moderate risk of bias, or serious risk of bias when a study has at least one serious risk of bias in the domains but no critical risk of bias [41]. Due to high heterogeneity in methodological approach and outcomes certainty of evidence using GRADE framework was not applied. The risk of bias assessment was carried out by the first author (UN), and the quality of the process was independently checked by the second author (II) with any discrepancies being discussed and resolved.

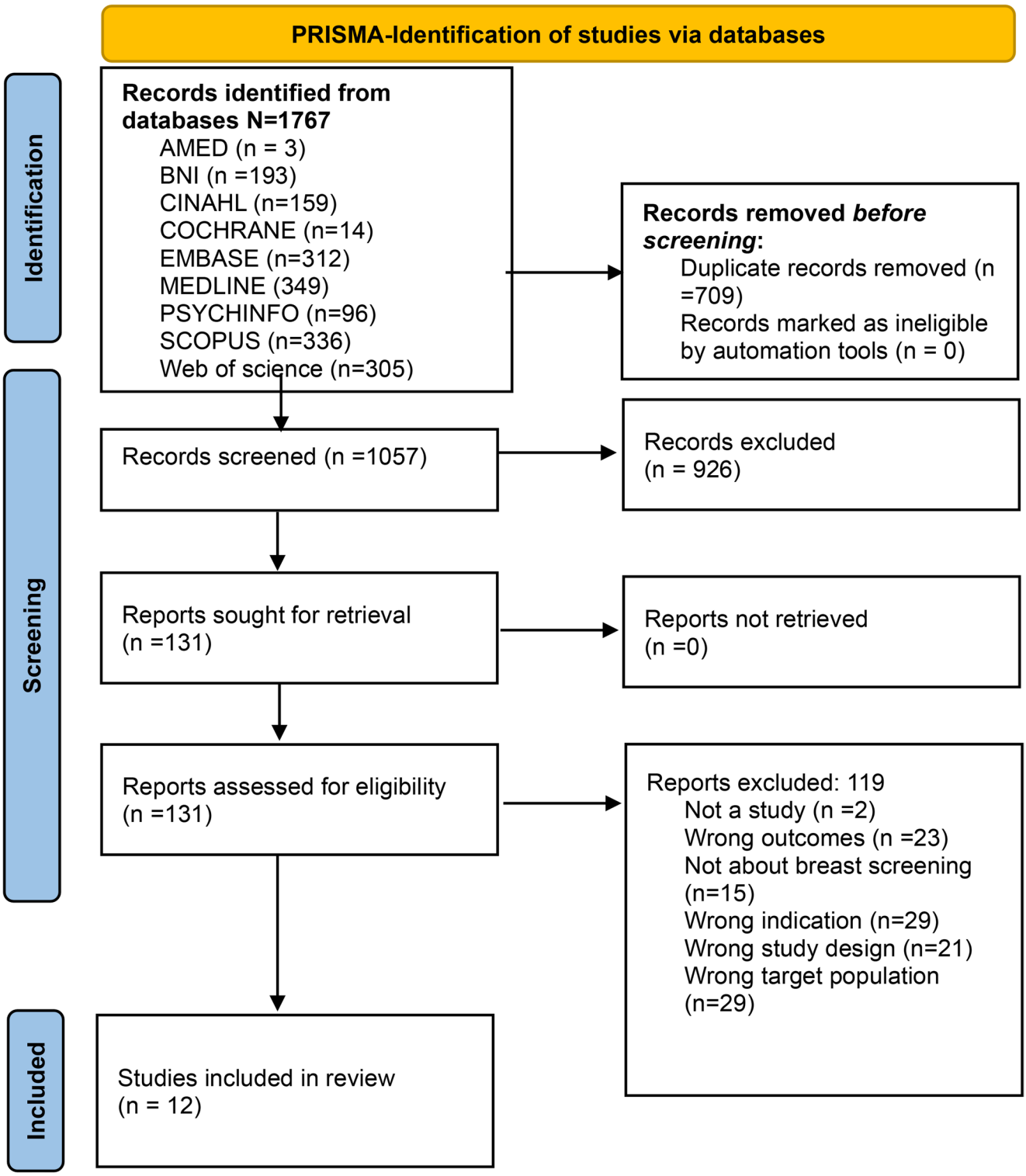


Fig. 1 Showing PRISMA diagram

Data Synthesis

There was clinical and methodological heterogeneity due to diversity in interventions, outcomes and study designs; so meta-analysis was not possible, narrative synthesis approach was deemed more suitable. A narrative synthesis was adopted using a textual approach to summarise the data using direction of effect plots as described by Cochrane Handbook [42–44]. The type and components of the intervention were extracted. The interventions were mapped to the BCT by analysing the details of the interventions and comparing it with the BCT descriptions as provided by Michie and colleagues [34], in order to explore patterns, differences/similarities between the interventions.

Effectiveness of breast screening interventions was analysed by assessing and re-calculating change in mammogram uptake, change in screening knowledge or intention to take up mammogram. Positive change is denoted by an upward triangle ▲; negative change is denoted by downward triangle ▲ and sideways arrows denote no change ⟷. Magnitude of effectiveness was analysed by the difference between the intervention and control or between post-test interventions versus pre-test interventions i.e. noting the proportional difference between the intervention and control as used by other studies [21, 45]. A small change of less than 5% is denoted by a small arrow, a change of 5-30% is denoted by medium arrow and a large arrow denotes a change greater than 30%. Descriptive statistics, inferential statistics with its confidence interval or p-values if available were retrieved from the studies. Binomial test for a sign test (two tailed p-value) using https://www.graphpad.com/quickcalcs/binomial1/ was applied. Here number of "successes" was defined as interventions whose outcome was 5% or above the baseline or reported as statistical significance.

Results

Risk of Bias Results

Figures 2 and 3 highlights the risk of bias for RCTs and for other non-randomised control trial studies respectively. Overall, two out of three RCTs had low risk of bias while one was classed with some concerns. Six of the non-randomised trials were classed as moderate concerns due to their process of participant selection while the other three had serious concerns. Although studies with serious concerns were identified, they were included in this review to capture the full scope of available evidence, providing valuable insights into the types of interventions that has been developed for Black African women. One study had a more rigorous selection procedure, with randomization in the first phase and so the classification for selection bias is low risk [46].

Overview of the Studies

The 12 studies on breast screening interventions for Black African women were all quantitative studies and of different designs: Randomised Control Trial (RCT) ($n=3$), quasi-experimental ($n=5$) and cross-sectional ($n=4$). All the studies were conducted in the United States. The review covered more than 4333 participants who were Black African women. One study using mass media did not provide number of participants [47].

The studies using RCTs assigned participants into the intervention or control group [48–50]; quasi experimental studies had pre- and post-intervention outcomes [46, 51–54] and other studies explored the number of participants who benefited from the intervention [55–58]. Eight studies adopted theoretical frameworks; elaboration likelihood of persuasion [48]; the Health belief model [49, 51]; transtheoretical model [46, 49]; persuasive health message framework [52]; social

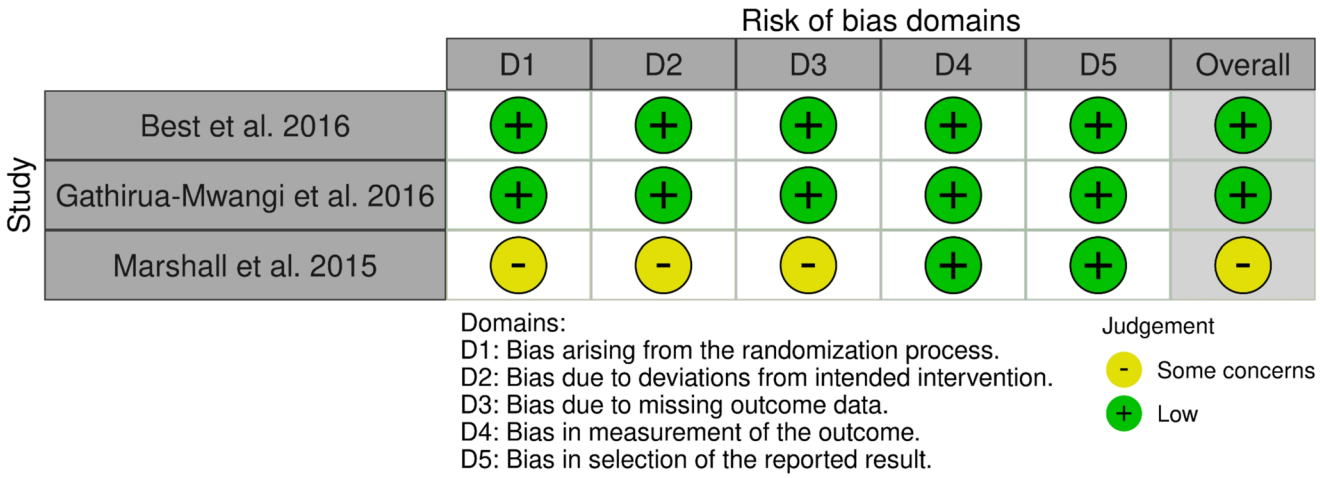


Fig. 2 showing risk of bias for randomised control trials

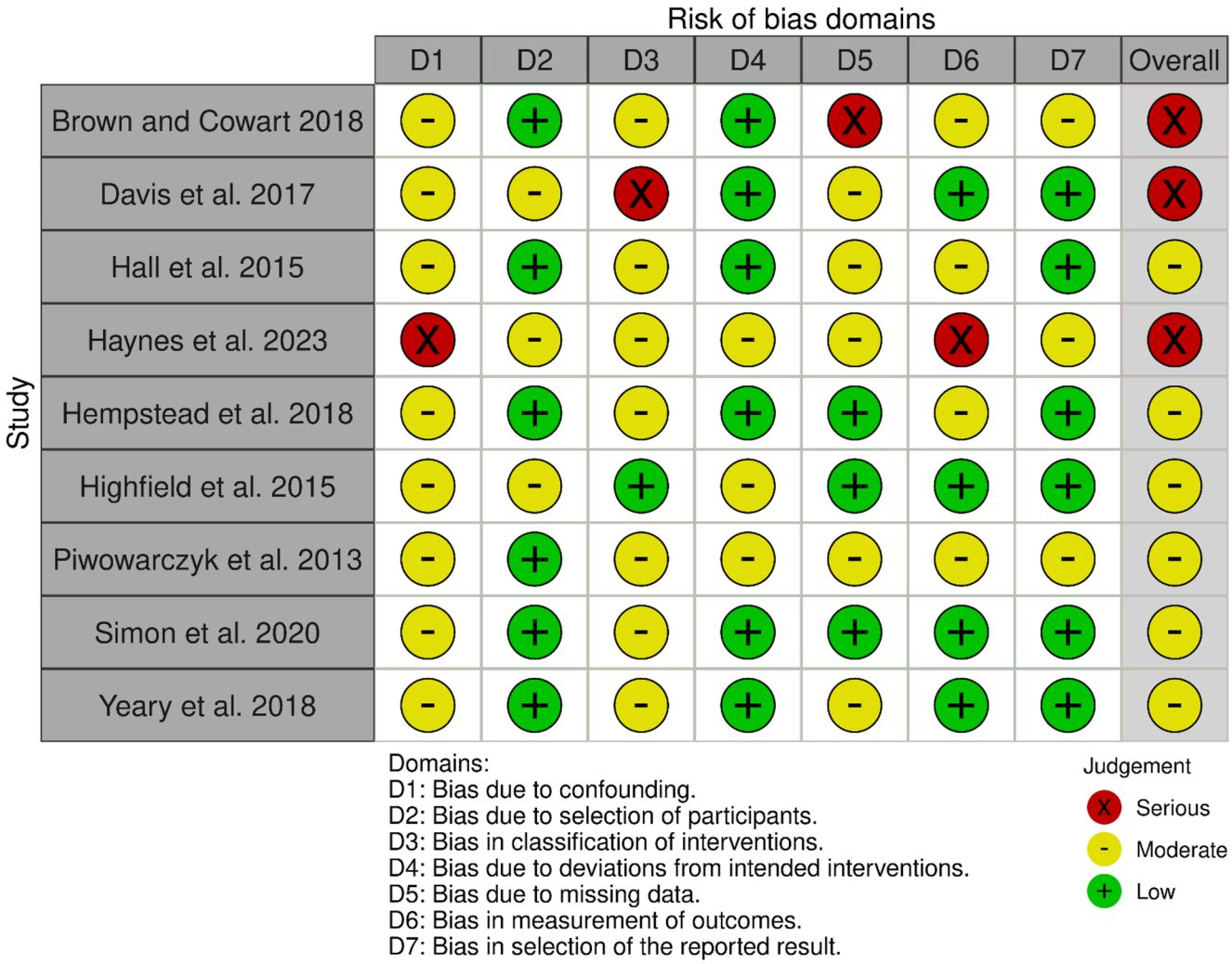


Fig. 3 showing risk of bias for non-randomised control trials

ecological model of health promotion [58] the outreach navigation Helping her live framework and the theory of scaling out [55]; social cognitive theory [56].

Description of the Intervention

Methods of delivery of the interventions varied across studies. Eight studies delivered the intervention by a single strategy using spiritual messaging [48]; telephone counselling or DVD presentation [49]; community educational workshop [53, 54, 56]; telephone counselling reminder [46]; Patient navigation [55]; breast health education [51]. Four studies reported using a combination of strategies. Hall and colleagues combined radio and print media to evaluate outcomes of dissemination of health promotion messages in different geographical locations (Savannah, Macon and Columbus) [52]. The radio station was selected due to their broad-African listenership while the print media were placed in community venues. While Haynes et al. employed community champions to deliver multiple events like faith-based activities, mammogram event, distribution of print flyers or had conversations with women [57]. Another study used a program which consisted of a 45-minute play, storytelling, skits about breast cancer using community members [58]. Printed educational materials and patient navigation were compared against printed educational materials only [50]. The types of interventions developed for Black African women to increase breast screening were grouped into (i) individual support/patient navigation (ii) health promotion or (iii) community health education programs. The components of interventions are summarised in the supplementary files and linked to behaviour change technique.

Outcomes of Interventions

The studies in this review had three main outcomes namely increasing breast cancer awareness/knowledge, intention to take up mammography and mammography uptake.

Eight studies showed more than a 30% increase in their outcome of either mammogram uptake ($n=5$), intention to take up mammogram ($n=2$), increase knowledge ($n=1$) as illustrated in the effect direction plot Table 2 below. These studies found their interventions were more effective than its comparator or when compared with before the intervention was delivered. There was evidence that the interventions influenced the individual barriers to screening. 11 out of 12 studies were effective, that is at least 5% and above increase in outcome favouring the interventions, with the binomial test showing two tail test P value 0.0063, using a significance level of 0.05. Details of the interventions mapped to the BCT are presented in the supplementary files. The study by Gathirua-Mwangi and colleagues had a less than 5% increase, there was also no statistically significance difference between the intervention and control [49]. The differences between the intervention and control were discussed below using percentages and odds ratio (Table 3).

Breast Cancer Screening Uptake

Seven studies used a range of interventions with the aim to increase breast screening uptake [46, 49, 50, 52, 55–57]. Studies that used individual support in the form patient navigation services, emerged as the most consistent effective strategy. Interventions using patient navigation as a single strategy or when combined with another strategy had more than 30% increase in breast screening uptake. For instance, when patient navigation services and printed educational materials (intervention group) was compared against printed educational materials only (control group), in an RCT, there was a statistically significant increase in mammogram uptake in the intervention group (93.3% $N=638$) versus those in control group (87.5% $N=720$) ($p<0.001$) and significant increase in odds (OR 3.63, 95% CI 2.10–6.26) [50]. The effect of patient navigation appeared more pronounced in participants who were overdue for screening. Of the small number of participants overdue for screening 182/1358 participants, a further analysis showed 73.4% increase with patient navigation and printed educational material group while there was 45.6% increase in breast screening uptake for those given printed materials only [50]. Similar quasi-experimental evidence using patient navigation as a single strategy, reported similar adjusted odds were obtained in the patient navigation group OR 3.88 times higher ($p<0.001$) than the control group [46]. In addition there was a 57% reduction in non-attendance of breast screening (calculated as percentage change) [46]. Similarly, this large impact in mammogram uptake were obtained amongst those who sought patient navigation and outreach services [55]. Although, these studies have different study designs namely-RCT [50], quasi-experimental [46]and cross-sectional [55] study design, evidence from these studies suggest a substantial positive effect but, these findings should be interpreted with some discretion due to moderate concerns associated with the risk of bias assessment.

In contrast, studies with health promotional message interventions, had mixed and less consistent results. Two lower quality studies-quasi-experimental and cross-sectional study design [52, 56], showed increases of 30% or more with breast screening uptake. While the other study [49] which is an RCT, with low risk of bias, showed less than 5% increase in breast screening uptake. This discrepancy in outcome suggests that the apparent effectiveness may have been influenced by study design/risk of bias or may be due the interventions used. For example, in the RCT, health promotional messages were delivered with DVD or restricted telephone counselling, neither DVD nor telephone counselling intervention were more effective than usual care; 41% for DVD, 42% for telephone counselling 35% for usual care ($P=0.6491$) level of significance 0.05 [49]. However, in the quasi-experimental study, health promotional messages in the form of print materials and radio messages had a large impact on breast screening uptake in different US cities with 46% and 20% increase respectively [52]. These were both higher than the comparison site at Columbus city at 15.3%. While community-based interventions using multiple strategies like breast cancer community champions, multiple awareness and screening events had a moderate increase in screening uptake [57]. The results from the radio or print media interventions, might have been influenced by other confounding factors especially if there is another campaign going on in the same area during that time [52].

In general, the evidence suggests that patient navigation interventions, offering personalised support are more effective for increasing breast screening uptake than information-based strategies. Although, the strength of this evidence might have been tempered by the methodological approach used in the studies.

Intention to Take Up Mammogram

There were three studies that evaluated intention to take up mammogram using-community-based program [54, 58] and health promotional strategies [48]. Overall, these studies that analysed mammogram intention, had a moderate to large impact on mammogram intention (i.e. from 5% and above). But the strength of evidence ranged from low to

Table 2 Showing study characteristics on breast screening interventions

Author year Country	Study aims	Sample Characteristics	Study Characteristics	Control description	Intervention(s)	Difference in attendance compared to control (%)	Effect direction and magnitude of effectiveness
Davis et al. (2017) [58] Country: USA	The development implementation, and evaluation of a community-based participatory research (CBPR) program designed to increase breast cancer screening awareness.	*N* = 357 Age range 40 years and above. Average age 57.38	Study design: cross-sectional Sampling: Purposive Duration: 2-year study period Follow up: 6 months Data Collection: pre/post questionnaire	N/A	Education intervention to dispel myths. 45-minute play, using community members and storytelling	Pre-test: 88.36% likely to schedule mammogram next year Post-test: 96.26% said they will likely schedule mammogram next year. Significant differences were indicated about pretest-post-test concerns about • costs • radiation • fear of finding cancer	13% difference in mammogram intent between pre and post intervention.
Hall et al. (2015) [47] Country: USA	To evaluate whether a culturally appropriate campaign using "Black radio" and print media increased awareness and utilization of local mammography screening services provided by the Centres for Disease Control and Prevention's National Breast and Cervical Cancer Early detection Program among African American women.	Implementation of campaign activities in two intervention sites in 1) Savannah with radio and print media 2) Macon with radio only 3) comparison site (Columbus, GA)-no intervention	The evaluation used a quasi-experimental design involving data collection during and after campaign Measured by an increase in phone calls to the Cancer In formation Service line, by uninsured, African American women aged 40–64 in the study sites compared to the comparison location during the campaign implementation period	P1 = period1(8/08–1/09) Active campaign and building awareness P2 = period2(2/09–8/09) P3 = period3(9/09–12/09) post-campaign period	Mass media black radio messages with facts.	Savannah (radio and print material). Change from P1 to P2 46% increase in mammogram. From P1 to P3 20% increase. Macon (radio only) Change from P1 to P2 20% increase. Columbus (comparison site) From P1 to P3 14.2% decrease in uptake Change from P1 to P2 15.3% increase From P1 to P3 17.9% increase	111% difference between intervention and control in the first period and 158% difference in screening in the 2nd period.
Gathirua-Mwangi et al. (2016) [49] Country: USA	To compare the effects of two interventions with usual care on mammography adherence among African American women	*N* = 244 Age range: 41–65 years AA women who had not had a mammogram in the last 15 months and no history of breast cancer	Study design: Randomised control trial. Method of data collection: self-report interview data and medical records of mammogram uptake.	Usual care which varied depending on location but ranged from nothing to a postcard reminder when it was time to schedule a mammogram	(1) Mailed interactive DVD. (2) computer-tailored telephone counselling	Overall mammography uptake increased similarly among all three groups: 41% for DVD 42% for telephone counselling 35% for usual care ($P = 0.6491$). Adjusted odds showed that neither the DVD (OR = 1.64;95%CI 0.80–3.39) nor telephone (OR = 1.24; 95% CI 0.61–2.50) intervention was more efficacious when compared to usual care. The DVD intervention was more effective than usual care for African American women with the lowest household incomes (<$30,000), but not for those with higher household incomes OR = 5.34 (1.12–25.42)	2.44% difference in screening uptake between intervention and control.

Table 2 (continued)

Author year Country	Study aims	Sample Characteristics	Study Characteristics	Control description	Intervention(s)	Difference in attendance compared to control (%)	Effect direction and magnitude of effectiveness
Haynes et al. (2023) [57] Country: USA	To improve breast cancer screening rates for women of African heritage through peer-to-peer education, which has proven successful for addressing cancer-related health disparities	12 Champions of African ancestry who conducted 245 educational activities and/or events in 15 months, of which 48 were virtual or phone-based and 197 were in-person.	Study design: cohort study 15 months. The analyses were supported using two datasets: 1) primary data collection of Champion's activities. Data on mammogram uptake was collected through medical billing information.	Area of no champion activity	Champion activities: Mobile mammography event Education event Faith-based	$N=405$ for intervention versus $N=360$ for control period. The proportion of residents participating in screening was significantly different between control and intervention time periods as determined by a Chi-squared test (X2 = 3.0845, $p=0.079$)	13% difference in screening rate between intervention period and control period.
Hempstead et al. (2018) [53] Country: USA	An examination of a peer educator intervention; in increasing breast cancer knowledge among African American peer educators to examine if workshop participants intended to share information, they learned from peer educators.	14 peer educators and 121 community members with 94 being African American. Age: 40 years and older	Study design: Quasi experimental design. Data collected were workshop pre-test and post-test. 14 peer educators trained as Community Empowerment Partners (CEPs) were breast cancer survivors (57%). Who in turn trained 94 women from the community.		1-day (6 h) training workshop	CEPs: Mean Pre intervention 6.25 (0.29) Post intervention 7.50 (0.45) (95% CI − 0.31, − 0.06) P value 0.005 Workshop Participants: Pre intervention 6.40 (0.15) Post intervention 7.38 (0.15) (95% CI − 1.09, − 0.30) P Value 0.001 Pre-workshop, 69.1% $n=65$ said they will talk to family and friends about breast cancer screening. Post-workshop 88% ($n=83$) indicated they would talk to family about breast cancer screenings. 89% ($n=84$) indicated they would talk to friends about breast cancer screenings. 82% ($n=77$) indicated they would talk to acquaintances about mammograms for breast cancer screening. Change in knowledge was associated with significantly greater odds of intending to talk about breast cancer screening with family, friends, and acquaintances.	20% difference between pre and post intervention.

Table 2 (continued)

Author year Country	Study aims	Sample Characteristics	Study Characteristics	Control description	Intervention(s)	Difference in attendance compared to control (%)	Effect direction and magnitude of effectiveness
Highfield et al. (2015) [46] Country: USA	To evaluate the effectiveness of an adapted mammography evidence-based Intervention (EBI) in improving appointment keeping for mammography in African American women	Intervention group ($N=151$) Control group ($N=198$) African American female age range: 36–64 years. Uninsured, income of ≤200% of the federal poverty level and an upcoming appointment for a mobile screening mammogram at a program partner site	Study design: A Quasi experimental Trans-theoretical Model Data was collected using databases. Duration: February to April 2012	Telephone reminder with scan details and location.	The intervention is a tailored telephone counselling 6–10 min reminder based on the Trans-theoretical Model of Change. This involves assessing a woman's stage of readiness to attend her appointment through a series of survey questions and counselled her through barriers to attendance.	The adjusted odds of a woman in the intervention group attending her appointment were 3.88 as compared to the control group ($p < 0.001$).	44.64% difference in mammography uptake between the intervention and control
Marshall et al. (2016) [50] Country: USA	To evaluate the effect of patient navigation on screening mammography among African American female Medicare beneficiaries in Baltimore, MD.	$N=1358$ African American female medicare beneficiaries who were ≥65 years of age	Study design: A randomized controlled trial conducted from April 2006 through December 2010. Setting: Community-based and clinical setting. Data collected was self-reported receipt of mammogram at the exit interview,	$N=720$ Printed educational materials on cancer screening.	$N=638$ Printed educational materials + patient navigation services designed to help participants overcome barriers to cancer screening	A greater proportion of participants receiving the patient navigation intervention reported getting a mammogram than those in the control group (93.3% and 87.5%, respectively; $p<0.001$). For women not screening-adherent at baseline, the incidence of mammography screening at study exit was 73.4% for those in the intervention group, compared to only 45.6% for those in the control group ($p<0.001$).	60.96% difference in screening for women who were not screening adherent at baseline. And 6.63% increase in screening for the target population.

Table 2 (continued)

Author year Country	Study aims	Sample Characteristics	Study Characteristics	Control description	Intervention(s)	Difference in attendance compared to control (%)	Effect direction and magnitude of effectiveness
Piwowarczyk et al. (2013) [54] Country: USA	To evaluate the UJAMBO program addressing the impact on participants 'knowledge of health services and intentions to use these services.	*N* = 120 African American women from Congo and Somali. Age range: 25–60 years	Study design: A Quasi experimental design using Pre-workshop post-workshop tests. Analysed using descriptive statistics and McNemar tests.		UJAMBO involved a one-session group workshop built around a DVD providing: a) basic information about these health services. b) normative change regarding their use through stories of African immigrant women's experiences with these services.	Knowledge of mammograph Pre-workshop: *n* = 77 (67%) Post-workshop: *n* = 109 (94.8%) Mammogram intention likely: Pre-workshop 58.8% Post-workshop 84.9% ($P < 0.001$).	42% increase in knowledge between pre and post intervention. 44% difference in mammogram intent between pre and post intervention.
Best et al. (2016) [48] Country: USA	To examine whether spiritual framing of BCS messages is associated with increased intention to obtain or recommend a mammogram (i.e. mammography intentions—primary outcome)	Inclusion criteria: a) self-identification as an African American woman. b) age 25 years and older. c) able to read, write, and understand English.	Study design: Randomised controlled trial. Duration: June and September 2012 Method of data collection: 7-point Likert-type scale survey ranging from very likely (7) to not likely at all (1). A total score (range = 4–28)	Traditional message *n* = 99 this was an 8.5 × 11" print-based, colour document that contained an image of three African American women of various ages. The top banner of the message contained the text "You can't put a price on life"	Spiritual message *n* = 100 It was developed by adapting the traditional message in consultation with African American women to incorporate spiritual elements	Participants that had spiritually framed message reported significantly greater elaboration (M = 5.35, SD = 2.35) compared to Traditional group (M = 3.97, SD = 2.58; $p < 0.001$). Participants in the spiritual group reported significantly greater polarity (M = 0.89, SD = 0.25) compared to the traditional group (M = 0.72, SD = 0.35; $p < 0.001$). Spiritual framing had a significant indirect effect on mammography intentions through polarity (standardized indirect effect = 0.057, 95% confidence interval [0.024, 0.106], $p < 0.001$)	

Table 2 (continued)

Author year Country	Study aims	Sample Characteristics	Study Characteristics	Control description	Intervention(s)	Difference in attendance compared to control (%)	Effect direction and magnitude of effectiveness
Brown and Cowart (2018) [51] Country: USA	To increase community engagement and collaboration concerning breast cancer awareness and education by engaging pastors, congregants and community members in educational sessions and conversations about breast cancer.	*N*=110 Age range: 13–86 years Average age 54.89	Study design: A Quasi experimental study. 6-month follow-up Method of data collection: questionnaire pre- and post-event surveys		Faith-based education	Increase in general breast cancer knowledge Mean change 0.881 (SD=2.128; 95% CI 0.184 0.517 1.244). At post-test, 76% of participants (*n*=91) attending the church-based seminars could name a local organisation providing access to free or low-cost mammograms, a 65% increase from pre-test, when only 46% (*n*=55) could identify such an organisation. ($p<0.000$)	A minimum of 20% increase in knowledge between pre and post intervention for African American population group who attended church-based seminars.
Simon, O'Brian (2020) [55] Country: USA	To evaluate Helping her live program implementation on the southwest side of Chicago to increase mammogram	West *N*=318(47%) Southwest *N*=94(16%)	Cohort study design. Method of data collection: records		The Community health workers (CHW) hosted workshops, attended health fairs, and conducted woman-to woman outreach activities, the CHW determines if the client would like help obtaining a mammogram or would like to receive a mammogram reminder in the mail. Clients who requested assistance with scheduling a mammogram were assigned to a CHW for breast imaging navigation.	Mammogram uptake in West *N*=110(32%) Mammogram uptake in Southwest *N*=26(8%)	66.99% difference in mammography uptake between intervention and control

Table 2 (continued)

Author year Country	Study aims	Sample Characteristics	Study Characteristics	Control description	Intervention(s)	Difference in attendance compared to control (%)	Effect direction and magnitude of effectiveness
Yeary et al. (2018) [56] Country: USA	To increase African American participation in breast cancer screening by embedding two approaches (comprehensive, focused) research in a state wide service program.	*N*=48 was African American Women. Focused *N*=19 Comprehensive *N*=29 Mean age of 63 years.	Cross sectional study design. Data collected surveys	Comprehensive group-addressed behavioural and biological risk factors associated with diabetes, heart dis ease and breast cancer.	Focused group-an oral presentation on the use of screening mammography, important strategies to reduce risk for breast cancer. Role models talked about them cancer experience in a culturally appropriate context. Lay advisors presented facts about breast cancer and informed participants about available cancer screening resources	3(18.75) out of 29 participants in comprehensive group had screening. 9(64.29%) out of 19 participants in focused group had screening	242.86% difference in breast screening uptake between intervention and control.

Table 3 Showing effect direction plot recalculated to show the difference between interventions and control or baseline

Study	Types of interventions	Study Design	BCT mapping	Number of strategies	Mammogram uptake	Mammogram Intention	Knowledge	ROB/ROB-INS
Simon et al. [55]	Individual support	Cross sectional	Problem solving Action planning Social support (practical)	Combination	▲			Some Concerns/ moderate concerns
Highfield et al. [46]	Individual support	Quasi-experimental	Problem solving Action planning Prompts/cues Social support (practical) Information about health consequence Verbal persuasion about capability	Single	▲			Some Concerns/ moderate concerns
Marshall et al. [50]	Individual support	RCT	Problem solving Action planning Social support (practical) Instructions on how to perform a behaviour.	Combination	▲			Some Concerns/ moderate concerns
Best et al. [48]	Health promotional messages	RCT	Prompts/cues Persuasive argument Information about health consequences Social support (emotional)	Single		▲		Low
Gathirua-Mwangi et al. [49]	Health promotional messages	RCT	Prompts/cues Persuasive argument Information about health consequences	Single	▲			Low
Hall et al. [47]	Health promotional messages	Quasi-experimental	Prompts/cues Persuasive argument Information about health consequences	Combination	▲			Some Concerns/ moderate concerns
Yeary et al. [56]	Community based workshops/program	Cross sectional	Prompts/cues Verbal persuasion about capability Social support	Single	▲			Some Concerns/ moderate concerns

Table 3 (continued)

Study	Types of interventions	Study Design	BCT mapping	Number of strategies	Mammogram uptake	Mammogram Intention	Knowledge	ROB/ROB-INS
Haynes et al. [57]	Community based workshops/program	Cohort	Prompts/cues Persuasive argument Social support (practical)	Combination	▲			Serious concerns
Brown and Cowart [51]	Community based workshops/program	Quasi-experimental	Prompts/cues Persuasive argument Information about health consequences Social support (emotional) Instruction on how to perform the behaviour	Single			▲	Serious concerns
Davis et al. [58]	Community based workshops/program	Cross sectional	Prompts/cues Persuasive argument Information about health consequences	Combination		▲		Serious concerns
Hempstead et al. [53]	Community based workshops/program	Quasi-experimental	Prompts/cues Verbal Persuasive argument Social support (practical)	Single			▲	Some Concerns/moderate concerns
Piwowarczyk et al. [54]	Community based workshops/program	Quasi-experimental	Prompts/cues Demonstration of behaviour Information about health consequences Social support (group setting) Verbal persuasion	Single		▲	▲	Some Concerns/moderate concerns

Small arrow < 5% increase in screening rate; ▲ medium arrow 5–30% increase in outcome; ▲ large arrow > 30% increase in outcome

serious risk of bias. In this case, community programs using culturally tailored approaches, had the largest effect, i.e., 30% and above.

For instance, community workshops using African immigrant women's stories distributed via DVD and guidebook, amongst Congolese and Somali women in the US, had a substantial increase in the intention to screen from 58.8% to 84.9% significant at $p < 0.001$ ($N = 120$) [54]. This was accompanied by a significant reduction in proportion of participants; from 18.5% to 2.5% reporting, they were unlikely to take up breast cancer screening after the workshop. This single strategy had a large impact, while similar community interventions using storytelling and skits had similar outcomes on the intention to attend breast screening [58]. This intervention was also effective as 96.26% (335) participants indicated that they would likely book a mammogram within the next year, compared to 88.36% (296) prior to the program ($x2(1) = 17.455$, $p < 0.001$) [58].

In contrast, health promotional messages using spiritually framed messaging showed more nuanced effect when compared with traditional messaging. Spiritual messages were positively associated with greater elaboration or cognitive engagement and more positive attitude towards

screening, their impact on screening intention was indirect and modest [48]. When considered together, these findings suggest that culturally tailored community-based intervention, is likely to be more effective in influencing screening intentions than health promotional messages.

Breast Screening Knowledge

Three studies used community-based interventions to increase and evaluate breast screening knowledge, all of which reported significant improvements. These studies typically used educational programs delivered by community members to improve breast screening knowledge. All three studies had a significant impact on knowledge [51, 53, 54], suggesting that locally embedded interventions may be effective for knowledge dissemination; as they reported consistent positive findings, with greater than 30% increase in knowledge for two of the studies and 5–30% increase in knowledge for the other study.

For example, community workshops with the Ujambo program, increased knowledge of mammogram significantly from 77 (67%) to 109 (94.8%), $p<0.001$ [54]. Similar large impact was also seen when church-based workshops were used by Brown and Cowart [51], with a 65% increase in awareness on how to access free or low-cost mammograms and a mean increase in general breast cancer knowledge, mean change of 0.881 which was significant ($p=0.001$). An increase in knowledge was noted when community educators were used to provide health education [53]. In addition, 88% ($n=83$) of the participants said they would talk to family (69.1% before the intervention) while 89% ($n=84$) would talk to friends and 82% ($n=77$) acquaintances about the screening [53]. Though the evidence and the magnitude of its impact is limited by moderate to serious risk of bias, nonetheless, consistency of positive findings across the studies suggests that community-based education is a promising intervention for improving breast screening knowledge.

In addition, other outcomes were reported in relation to breast cancer awareness using community programs. Interventions that apply, culturally adapted approaches in the form of plays and skits had significant reduction in women's perception of barriers to screening like worry about costs of mammograms; radiation exposure from the screening; fear of finding cancer and myths that getting treated for breast cancer is worse than the disease (all $p<0.001$) [58]. These findings suggest that beyond improving knowledge and breast screening uptake, community-based interventions may help in addressing misconceptions and emotional barriers to screening.

Discussion

The aim of this systematic review was to identify and evaluate interventions developed for Black African women to increase breast screening in high-income countries. To the best of our knowledge, this is the first systematic review that has explored breast screening interventions and their impact on breast screening uptake in Black African women in high income countries. We found there were three main types of intervention: individual support, health promotion messages and community-based interventions. All but one study showed significant impact in either increasing knowledge about breast screening, breast screening uptake or intention to take up mammogram. The findings suggest that individual support or patient navigation is the most successful intervention for increasing breast screening uptake. Whilst community-based interventions using community education workshops, culturally adapted programs or church-based programs were effective in increasing breast screening knowledge. The studies also showed a significant improvement in intention to take up mammograms through educational programs, however, whether this intention led to uptake of screening was not followed through. It is important to understand the difficulties associated with behaviour change and strategies to encourage this change. Intention is a poor predictor of health behaviour change as they are factors which may prevent them to follow through such as the trigger, capacity, motivation, the response and the process [59].

The findings in this review show that patient navigation that align with problem solving and action planning BCT [34] has the potential to overcome some provider level barriers to obtaining mammogram and is the most promising intervention for increasing breast screening uptake. Acharya and colleagues also had similar outcomes with problem-solving BCT, establishing problem-solving intervention as the most effective [21]. Other interventions targeted mainly individual level barriers rather than reflecting the wider provider or system level barriers faced by Black African women. Patient navigation is defined as individualized assistance provided to patients, relatives or carers helps to overcome provider and health system barriers [60]. Though, the risk of bias for the studies in this review on patient navigation is moderate, it has also been shown in other studies to be effective across the cancer continuum, with good evidence in breast cancer screening and early detection in the general population [31]. This intervention not only seeks to improve knowledge, but patient navigators help to answer personal questions, help with scheduling convenient appointments and access. This review found that patient navigation had

a large impact (more than 30%) in breast screening rates in Black African women. Likewise, Genoff et al. reported significant increase in breast screening rates for women with limited English proficiency [28].

The findings show that, when patient navigation was combined with an outreach program or printed media there was significant increase in breast screening. In other studies, patient navigation was combined with tailored DVD and there was also a significant impact in uptake of breast screening in women who were not up to date with screening [61]. Similar effect was identified when patient navigation was combined with mass media or general education for general cancer screening services in the general populations and underserved populations [31]. As a single strategy in this review, patient navigation equally showed a large impact on breast screening uptake. Marked improvement in screening with patient navigation as a single strategy has also been reported in other studies [22, 31]. Therefore, patient navigation as a single strategy or combined with another strategy is effective in improving screening. This is reasonable as patient navigator works by understanding an individual's barriers helping them overcome these barriers. Therefore, public health policy should include the use of patient navigators to approach Black African women to improve screening.

This review found that community based educational programmes had significant impact in increasing knowledge. However, the risk of bias for these studies were moderate to high. The community-based interventions aligned with previous recommendations about the need to disseminate cancer screening messages through the community; faith settings like the church/mosque, along with tailoring health messages in the media to appeal to Black African community [62]. This favourable outcome was in line with other studies who analysed community based educational programs to improve breast screening awareness in other population group like in Arab women [63], ethnic minority women [27] and in the general population [64, 65]. When the effect of education on breast screening was analysed it showed that women's behaviour changed positively following educational programs with 28.4% of women receiving mammogram 6 months after the educational program [66]. In this review, the studies exploring community-based education showed a moderate to large increase in Knowledge scores. Therefore, the use of community based educational programs should be considered by public health practitioners to increase awareness or knowledge.

According to the Institute of medicine in the US, belief systems, cultural values and group identity are all powerful tools through which information is filtered and processed [67]. These tools were utilised in this review as faith-based health promotional program and culturally adapted community programs were found to have a positive effect on breast cancer screening awareness. Similarly, a study on culturally adapted health education program amongst immigrant and refugee women in the US, reported significant increase in knowledge scores, with 35% of their participants utilizing breast screening service during the follow-up period of the study [68]. Likewise, in amongst Chinese American immigrants in the US [69].

Similar post-intervention effects were obtained in this review with the use of belief systems as a tool for health education. In this review, faith-based messages generated positive thoughts about screening which increased mammogram intention. This is supported by Hou and Cao review in ethnic minority groups which revealed that faith-based cancer education interventions significantly improved knowledge [29]. Likewise, a Muslim-focused qualitative study in Scotland found that participants regarded screening messages in the context of Islam as good, with all participants reporting positive thoughts and feeling engaged with video messages of Muslim women addressing cancer screening [70]. Within the Christian community, similar acceptability of faith-based messages and the importance of pastoral input was highlighted in cervical cancer screening [71]. These suggest that knowledge improvement interventions will likely have more impact when cultural and faith-based values are considered and incorporated. Due to the acceptance of cultural and faith adapted interventions, future interventions should consider working with community stakeholders to incorporate relevant acceptable strategies.

In addition, this review, showed a significant increase in intention to attend breast screening, following a faith-based intervention and community based participatory intervention. Like Hou and Cao acknowledged, even though faith based cancer education improved knowledge it did not always increase screening uptake [29]. This may be due to the intention-behaviour gap. With breast screening educational workshops, knowledge and intention to screen may increase but it is possible that participants may not follow through after these interventions. The process of change in behaviour is complex and determined by various factors like motivation, capacity and the process [59]. Therefore, if the aim is to increase breast screening, obtaining mammogram intention feedback following an intervention may not be enough.

These findings have implications for public health nurse practitioners and policy makers to adopt an approach to screening services that incorporates belief systems, are culturally sensitive and to develop an adaptable health service in partnership with community members from diverse ethnic, religious and cultural backgrounds. This is important to foster trust and ownership in the health system. This review provides clarity

on the need to develop and explore interventions for improving breast screening for Black African women in High income countries which public health researchers could explore further. Considering that the study designs were heterogenic, studies should explore the difference between specified single and the use of multiple interventions in randomised control trials. More studies are needed to highlight the impact of interventions in specific ethnic groups in high income countries and to explore interventions to resolve provider level and health system barriers. These findings will be beneficial to charities or organisations tasked to improve early detection as it provides suggestions on different interventions that could be explored with Black African communities to improve breast screening.

Strengths and Limitations

This review is novel in its population specificity, structured behaviour change analysis, and quantitative direction-of-effect synthesis, which together offer a more precise understanding of which breast screening interventions work for Black African women in high-income contexts. One of the key strengths of this review is that it utilised various relevant databases with wide scope of international literature to retrieve articles. The year of inclusion was broad enough to highlight recent interventions, thus highlighting different interventions that has been developed for Black African communities. Although grey literature was excluded to ensure the quality of the included studies, in reality the quality varied. The studies had varied risk of bias with most of the studies having moderate risk of bias, in addition certainty of evidence using GRADE framework was not applied. Therefore, there are limitations to the interpretation of the impact of the interventions. All the studies were included as they provide valuable insights into the types of interventions that has been developed for Black African women, their exclusion will not highlight a complete picture of the different interventions. There was variation in methodology and outcomes in the studies, which made it difficult to analyse the effect of the interventions further.

Conclusion

Almost all the interventions in this review showed significant impact in either increasing knowledge, breast screening uptake or mammography intentions, though their impact varied. Thus, indicating that a variety of interventions could be effective in improving uptake in Black African women, giving individual support was the most impactful. Primary studies on breast screening interventions are required in other high-income countries that serve as host country for Black African women like the United Kingdom.

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Authors' contributions All authors contributed to the study conception and design. Ugomma Nwadinigwe conducted the searches, screened the articles, extracted the data, evaluated the study quality, mapped the results to Behaviour change technique and analysed the results. The quality of the search results, screening process, data extraction, quality appraisal of the studies was independently checked by the second author Imose Itua. Chris Keyworth independently reviewed the mapping of the results to the behaviour change technique. Mel Cooper reviewed the whole review process to ensure congruency from the study inclusion to outcomes and conclusion. All authors read and approved the final manuscript.

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Code Availability not applicable.

Data Availability This is a review of other studies. All data supporting the findings of this study are available within the paper and its Supplementary Information.

Declarations

Ethical approval This is a review of other studies; ethics is not applicable.

Consent to participate not applicable.

Consent to publish not applicable.

Clinical trial number not applicable.

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